PEPII

RF SYSTEM

RF NON-IONIZING RADIATION SAFETY PROCEDURE

PS-340-330-61-R2

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RF NIR SAFETY PROC

Stanford Linear Accelerator Center
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RF NON-IONIZING RADIATION SAFETY PROCEDURE FOR PEP-II RF STATIONS

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1) Introduction: (see sketches: Layout Region 8 Support Building and Typical Cross-Sectional Layout of RF Station)

The RF system of PEP-II consists of 5 stations in the high energy ring (HER) and 3 stations in the low energy ring (LER). A station has the following components of interest to this procedure: A 1.2 MW CW klystron with a 120 Watt drive amplifier; a circulator with a 1.2 MW load; a waveguide distribution network splitting the power four-ways (HER) and two-ways (LER) including one to three 1.2 MW loads; four or two penetration waveguides transmitting the RF into the tunnel; four (HER) or two (LER) cavities in the tunnel. The operating frequency is 476 MHz.

All RF fields are normally contained in metal enclosures like the klystron body, rectangular waveguide or resonant cavity. The klystron and RF cavity are vacuum vessels and do not leak RF fields if they are vacuum tight. Leakage out of the waveguide is normally limited to joints that are either defective, improperly assembled or insufficiently tightened.

A RF leak tight waveguide network is assured by taking the appropriate steps during assembly and installation and checking by inspectors after installation for proper assembly. A final RF survey using a calibrated radiation meter will be made for each joint accessible in the surface buildings. As an additional safety net most of the waveguide network is part of a pressurization system which will automatically shut off the RF power and abort the beam if the waveguide loses its pressurization due to a gas leak assumed to be accompanied by a RF leak.

2) Requirements:

A personnel hazard is considered to exist when RF radiation Non-ionizing Radiation) leaking from any source in the system exceeds a power density level of 1.5 mW/cm^2 at the surface (for operating frequency: 476 MHz).

Reference: American National Standard Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 300 kHz to 100 GHz. (ANSI C95.1-1982)

3) Non-ionizing Radiation Safety approach:

The radiation integrity of the power system of PEP-II RF stations is being assured by the following methods:

- Careful assembly of all parts followed by an inspection.
- RF field survey of the fully assembled system.
- Pressurization of waveguide network tied to interlock for pressure failure as additional safety net.

4) Waveguide Network: (see Rectangular WG Network Region 8 Subassembly Drwg # SA-341-300-05)

The waveguide network consists of WR2100 aluminum waveguide with two types of flange seals. In the penetrations PARKER SEALS are used to seal for RF and gas leakage. This sealing method was used in PEP-I successfully and is reused in the PEP-II penetrations where the old waveguides are being reused. The rest of the installation uses flat flanges paired up with grooved flanges containing Silicon Rubber O-rings.

Several parts of the waveguide network need special mention:

1) The klystron RF output beyond the vacuum window has a transition from a 6 inch coaxial line to WR2100 waveguide which is part of the pressurization system.
2) A 1.2 MW circulator was procured specified to be leak tight and is part of the pressurization system. A gas barrier on the downstream side of the circulator is inserted for protection of the circulator and is pressurized on both sides.

3) 1.2 MW loads are attached to circulator and magic tees through waveguide gas barriers. The loads have large water drain holes to prevent the waveguide network from filling with cooling water in case of a catastrophic load failure. The gas barrier is held in place between the load flange and the waveguide by long bolts. This joint is part of the pressurization system. Radiation safety will be assured by inspection and radiation survey.
4) The cavity window has a special flange at the ceramic vacuum window sealed with O-rings which is part of the pressurization system.
5) Infrared monitors measure the operating temperature of each cavity window through a monitor port. This port provides radiation shielding due to being beyond cutoff at the operating frequency. It will also be made leak tight for gas leaks.

5) Coaxial Cables:

The coaxial cables between the 120 W drive amplifier and the klystron have warning labels at each connector. The labels specify that there could be a power of up to 120 W present and that the cables should not be disconnected while power is on.

6) Assembly and Installation:

During assembly and installation of the waveguide components all flange bolts shall be torqued to 30 ft-lbs and all field assembled waveguide joints shall be tested by pressurization to 0.25 psig and checked for bubbles with "Snoop". After installation, the inspector shall measure the torque on a minimum of six bolts chosen at random on each flange. If the torque exceeds 25 ft-lbs the flange passes inspection, if not all the bolts on the flange are to be re-torqued to 30 ft-lbs.

7) RF Field Survey:

After the requirements for bolt torquing and gas leak checking are satisfied, a check for RF leakage around each accessible flange using a calibrated RF field meter shall be made with the klystron output at about 100 kW. The RF leakage shall nowhere exceed the prorated allowable limit of 0.1 mW/cm^2 at 100 kW. The coax connectors at the klystron drive amplifier, at the directional coupler and at the input to the klystron will also be surveyed for leakage at this time.

8) Gas Pressurization System:

Each waveguide network will be pressurized with regulated 0.25 psig instrument air which has been processed through an air drier. Since the volumetric supply rate is limited, a leak in the waveguide will cause a drop in pressure, actuation of a pressure switch and shut down of the RF station together with a beam abort.

After a leak is repaired a radiation measurement with the klystron operating at about a 100 kW power level is to be performed to check if the repair was successful. When the 0.25 psig pressure in the waveguide is restored and the RF radiation survey shows no leakage above 0.1 mW/cm^2 normal ring operation may resume.

Pressurization mainly guards against operation with a missing piece of waveguide or an improperly assembled flange joint. Proper torquing of all flanges to 30 ft-lbs is a necessity to prevent RF leakage at the waveguide flanges.

9) Klystron removal:

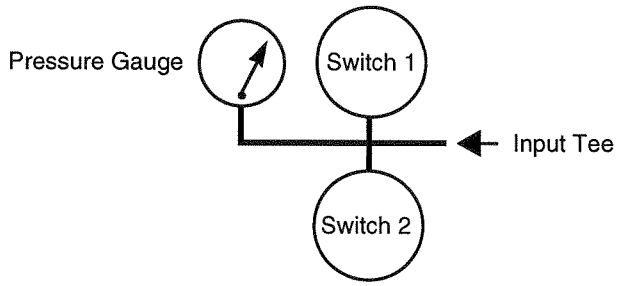
Before a klystron can be removed from the waveguide network the high voltage power supply has to be locked off using proper LOCK-AND-TAG procedures to prevent exposure to high voltage and accidental operation of the klystron.
If a klystron is to remain disconnected from the waveguide network it is required to bolt a cover over the open end of the waveguide. The cover will be properly torqued and inspected and the waveguide pressurized before any RF systems can be operated and a beam stored in the respective PEP ring.

10) Re-certification annually and after major repair or downtime:

RF radiation surveys shall be performed on all stations at least once per year and are the responsibility of the RF Group.
After each opening of a waveguide network or major downtime of the machine an inspection of the systems checking completeness and readiness for turn-on shall be performed including a RF radiation survey at 100 kW klystron power.
A RF radiation survey shall also be performed on all stations at full beam current to verify compliance with beam induced power at higher frequencies.

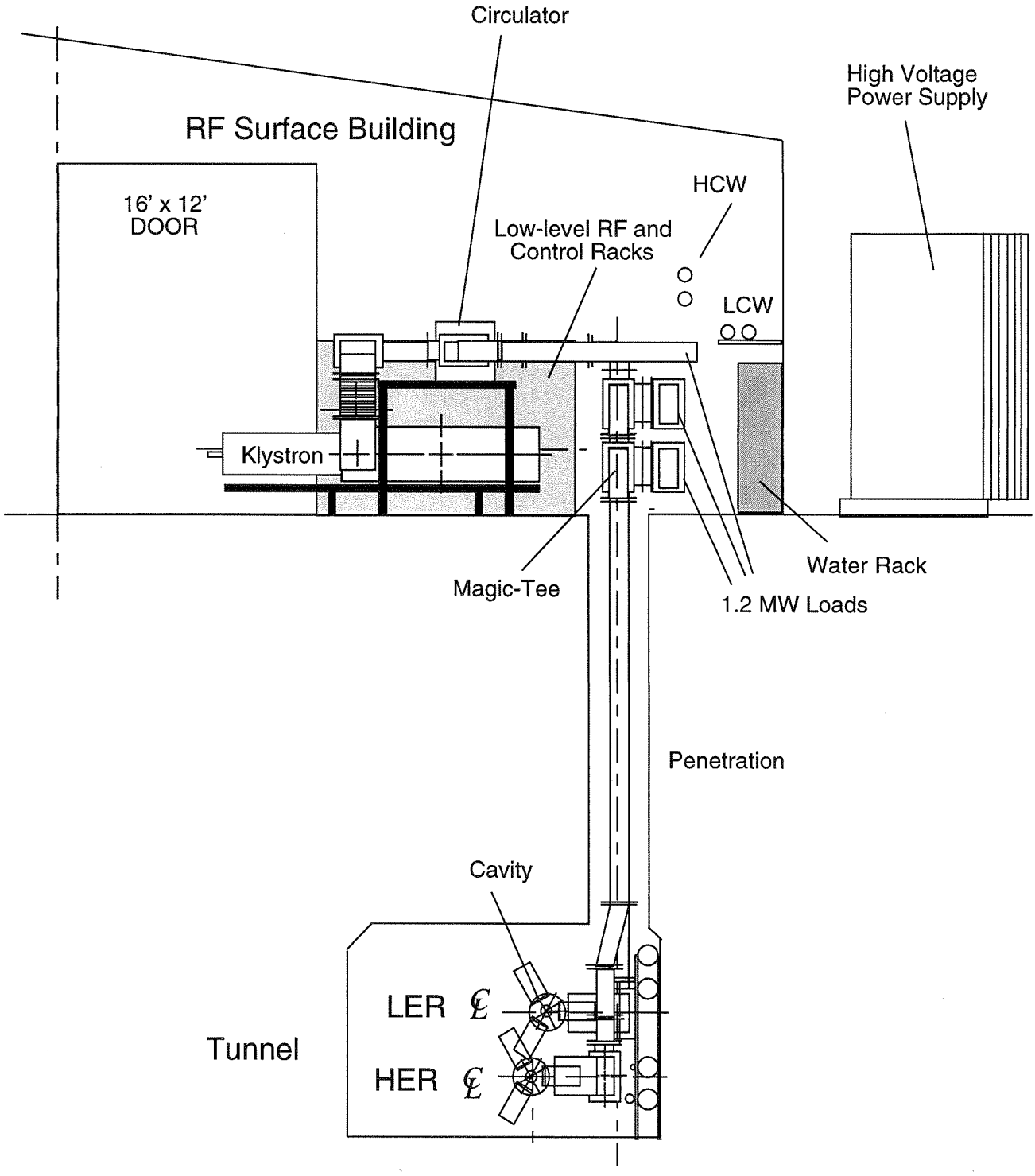
11) Waveguide Pressure Interlock Check:

Verification of waveguide pressure interlocks shall be performed on all stations at least once per year or after a major downtime. Proceed as follows:
Locate the waveguide pressure switches below the circulator loads. There are two sets of switches and one pressure gauge for each of the two zones. Locate the input air Tee to switches (see diagram below).

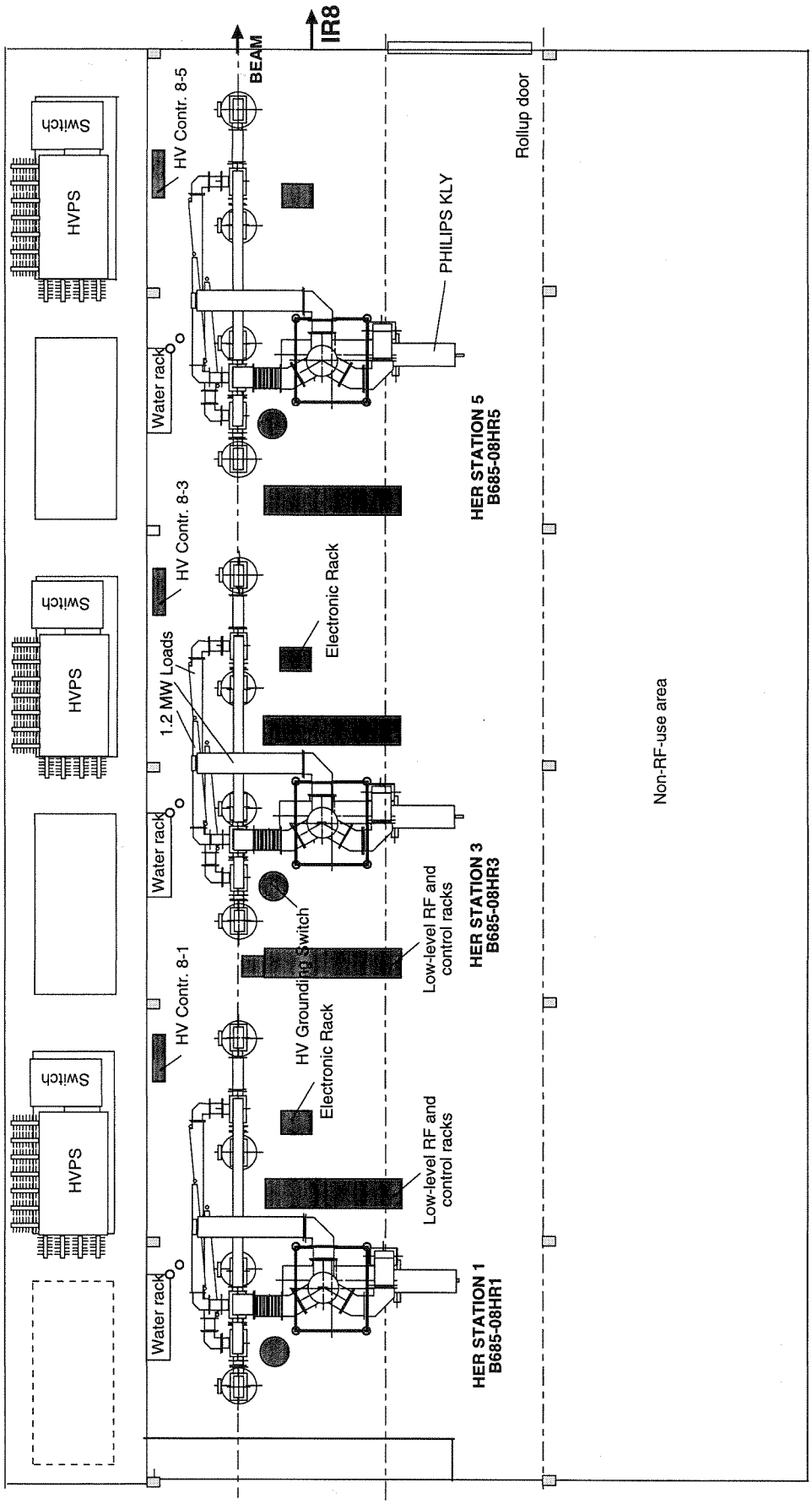


With a wrench, loosen the compression fitting on the 3/8 copper tubing supply to the Tee. Gently pull the tubing out of the compression fitting socket while watching the pressure gauge. Carefully dip the gauge pressure down to 3 inches. Verify Local Panel Waveguide Pressure LED has tripped to Red. Verify on the EPICS Panel (WAVEGUIDE) that both the Local Pressure Switch has tripped and the NIRP Output for that station has tripped. Hit Reset and repeat as necessary. When satisfied re-connect and re-tighten air supply.

Typical Cross-Sectional Layout of RF Station



Plan view of Region 8 RF Support Building B685



12) Waveguide Safety Work Control Procedure

For purposes of working on the PEPII waveguide system a *Waveguide Safety Work Control Form* (**WSWCF**) has been generated. There are two general occasions for usage:

I. First for any waveguide component removal, repair, klystron /circulator repair or removal. This requires PEPII Bend Magnet Chopper Supplies in Region 8 (Building 685) to be Locked Off. This prevents stored beam from inducing RF in the cavities that could propagate up the waveguide.

After component(s) has been removed and waveguide system shorting plates installed; if necessary, the waveguide air pressure system will make-up and PEPII Bend Magnet Chopper Supplies can be released- Lock & Tags removed.

1. Prepare *Waveguide Safety Work Control Form* for ADSO office (MCC).

2. PEPII Bend Magnet Chopper Supplies in Region 8 (Building 685) Locked Off by Person Responsible and people performing work- Lock & Tag. PEPII Area Manager or EOIC after hours will Lock Off Bend Supplies first. See Figure 1.

   LER Ring: Lock Off (+) and (-) Feeder Rectifier Supplies.
   HER Ring: Lock Off (+) and (-) Feeder Rectifier Supplies.

3. *Waveguide Safety Work Control Form* Signatures Required:
   Person Responsible (Engineer, RF Group)
   Area Manager (PEPII Area or PEPII RF Area)
   ADSO

4. Lock and Tag RF station HVPS off.

5. Requirements After Completing Work: W.G Pressure Relay Test. See section 11 of PS-340-330-61-R0. Close out WSWCF in ADSO office.

There are no sign-offs necessary for RHP, PPS Bypass in this situation.

II. Second for station commissioning or testing HVPS, klystron, or circulator that requires klystron full power without RF cavities. This requires installing a waveguide shorting plate on the circulator output. To run the station requires HVPS PPS Bypass and sign-off by RHP. This requires a *Radiation Safety Work Control Form(***RSWCF).**

1. Prepare *Waveguide Safety Work Control Form* and *Radiation Safety Work Control Form* for ADSO office.

2. PEPII Bend Magnet Chopper Supplies in Region 8 (Building 685) Locked Off by Person Responsible and people performing work- Lock & Tag. PEPII Area Manager or EOIC after hours will Lock Off Bend Supplies first. See Figure 1.

   LER Ring: Lock Off (+) and (-) Supplies.
   HER Ring: Lock Off (+) and (-) Supplies.

3. RHP or ADSO padlocks waveguide shorts in place and signs off *Radiation Safety Work Control Form* for HVPS PPS Bypass.

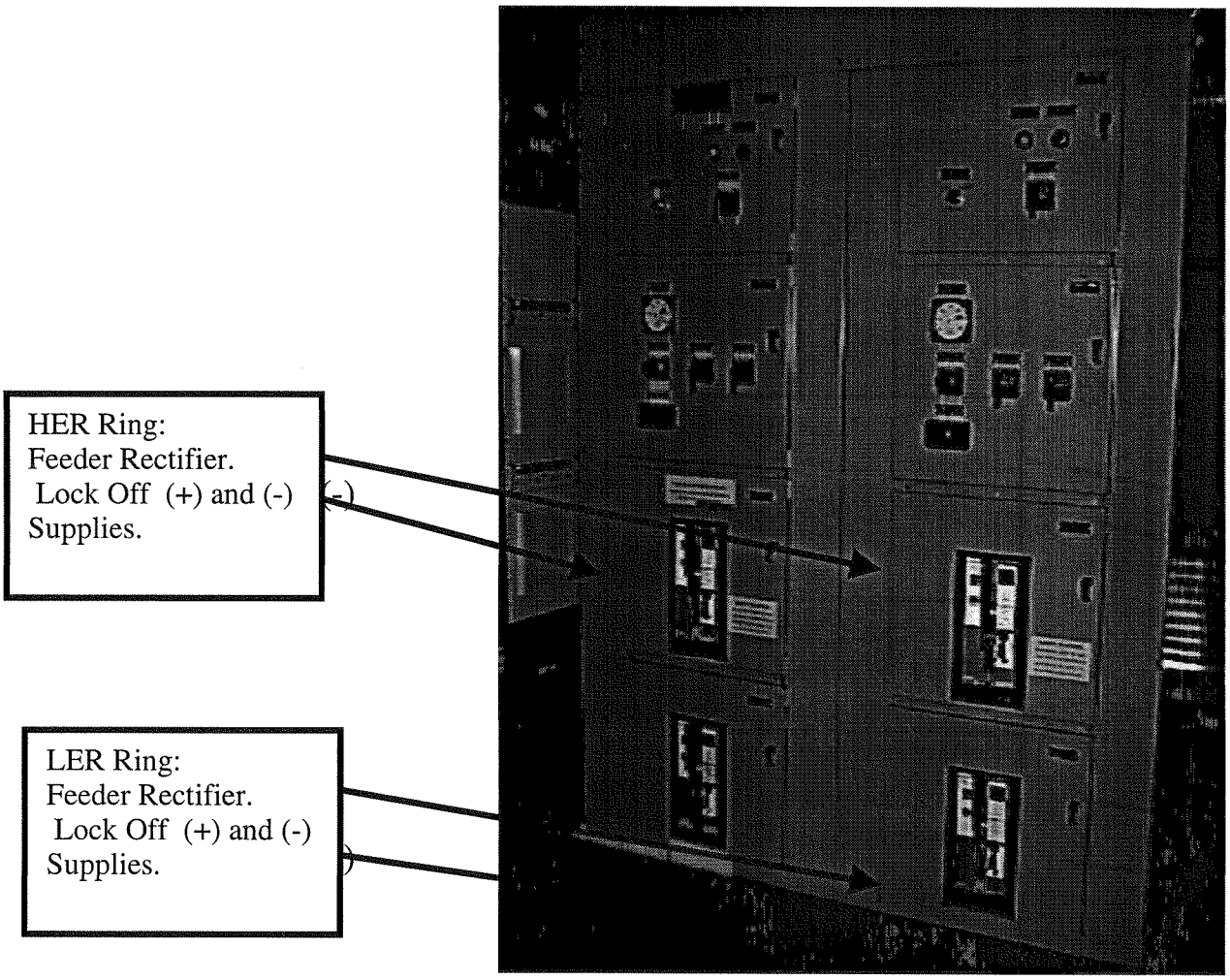
4. Have HVPS PPS system bypassed for station.

6. *Radiation Safety Work Control Form* Signatures Required:
   Person Responsible (Engineer, RF Group)
   Area Manager (PEPII Area or PEPII RF Area)
   ADSO
   RHP
   PPS

7. Remove waveguide bellows after circulator and install two waveguide shorting plates- one on each side of opening. Install waveguide short locking covers.

8. Requirements After Completing Work: W.G Pressure Relay Test. See section 11 of PS-340-330-61-R0. Close out WSWCF and RSWCF in ADSO office.

FIGURE 1. Located in Building 685 (Region 8 RF Support Building) Adjacent to Rack # B685-08CM25



Accelerator Department

PEPII RF Waveguide Safety Work Control Form

Area: Form #: Date:

Section 1: Description of Work to be Done (include item or CATER number, job title, etc.)

☐ Repair or Replace ☐ Install ☐ Remove ☐ Reinstall ☐ Bypass ☐ Unbypass

EXAMPLE

Person Responsible signature/date
(RF Engineer)
Mngr.)

Area Manager signature/date
(RF Area Mngr. or PEPII Area

Section 2a: Requirements Before Starting Work (ADSO completes this section)

Section 2b: Requirements After Completing Work (ADSO checks required boxes)

☐ W.G. Pressure Relay Test ☐ Operations ☐ Other (describe)

ADSO signature/date

Section 3: Signoffs Indicating Requirements are Complete (ADSO checks required boxes)
signature/date

☐ **Requirements Before Starting Work Completed** (See Section 2a):_____

☐ **Removed or Bypassed** (Operations, or Person Responsible): _____

☐ **Reinstalled or Unbypassed** (Operations, or Person Responsible): _____

☐ **Work complete** (Person Responsible or Area Manager): _____

☐ **Operations:** _____

☐ **Other:**_____

Section 4: Signoff Indicating Readiness for Beam to: _____

ADSO signature/date

EOIC signature/date